

State-Dependent Forecasting in Volatile Times^{*}

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Abstract

Recent inflationary episodes have highlighted the importance of understanding how volatility affects expectation formation. Using professional inflation forecasts, we document that forecast updating is state dependent: in volatile periods, revisions become more frequent and larger, forecast accuracy deteriorates, the influence of the consensus weakens, and disagreement rises sharply. To interpret these facts, we develop a Bayesian forecasting model with regime-switching inflation with three forecasting frictions: learning about the regime, revision costs and strategic concerns. Our findings show that these forecasting frictions are quantitatively important and that their interaction with inflation volatility is key to explaining why forecasts become more responsive in volatile times, yet not necessarily better informed. This has implications for the interpretation of inflation surveys and for monetary policy.

JEL: D80, D81, D83, D84, E20, E30

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1 Introduction

The recent surge in inflation has reignited interest in how inflation expectations evolve during periods of macroeconomic volatility. Central banks closely monitor expectations, often through surveys of professional forecasters, to gauge inflation persistence and guide monetary policy decisions. Yet volatile inflation changes the forecasting problem in fundamental ways: forecasters must judge not only where inflation is, but also whether it has become more persistent, whether the economy has shifted into a different regime, and how strongly to respond to signals that may differ from the consensus. At the same time, reported survey forecasts are shaped by motives other than accuracy: forecasters may avoid frequent revisions to preserve credibility, face institutional costs when rewriting narratives, and remain close to the consensus for strategic or reputational reasons. As a result, reported forecasts may differ systematically from underlying beliefs. A key and largely overlooked question is how inflation volatility interacts with these forces and reshapes the dynamics of reported forecasts.

We address this question by showing that macroeconomic volatility significantly alters the microeconomics of expectation formation. Using individual-level data from Bloomberg’s Economic Forecasts (ECFC) survey of professional forecasters, we document strong *state dependence* in forecast updating. In tranquil periods, when inflation is relatively stable, forecasts are lumpy and revised infrequently. In volatile periods, by contrast, revisions become both more frequent and larger, forecast accuracy deteriorates, the influence of the consensus weakens, and disagreement rises sharply. In other words, volatile times make expectations more responsive, but not necessarily better informed.

To interpret these facts, we extend the lumpy-forecast framework in [Baley and Turen \(2024\)](#) to allow for regime-dependent inflation dynamics and imperfect information about the underlying state of the economy. The model combines three key frictions. First, forecasters face a fixed cost of revising their reported forecast, which generates inaction and lumpiness in survey responses. Second, forecasters care about staying close to the consensus, capturing strategic or reputational concerns. Third, forecasters receive private signals about a latent inflation regime, so that agents differ in how they interpret incoming information and, therefore, in when and by how much they revise. Inflation itself follows a regime-switching process, so both the level of volatility and the uncertainty about the current regime shape forecast behavior.

A central implication of the model is the distinction between underlying beliefs and reported forecasts. Beliefs evolve as agents update using Bayes’ rule, combining observed inflation realizations with private signals to infer the latent regime. Reported forecasts, however, adjust only when the gain from revising exceeds the fixed revision cost. This wedge is essential for interpreting survey data: even when agents learn every period and their beliefs move continuously, reported forecasts may remain unchanged for long stretches of time. The model therefore provides a micro-founded mapping from latent beliefs to observed survey responses.

The model naturally generates state-dependent forecast dynamics. In tranquil periods, inflation is relatively stable and the low-volatility regime is likely to prevail. Beliefs then move gradually and often remain within the inaction region, so revisions are infrequent and typically small. In volatile periods, by contrast, inflation realizations become more informative about possible regime changes and posterior beliefs move more sharply. This raises the likelihood that beliefs cross the revision threshold, increasing the frequency of updates. At the same time, the greater uncertainty associated with turbulent periods raises the value of waiting, so that when forecasters do revise, they revise by larger amounts. The model thus generates the joint rise in the extensive and intensive margins of forecast adjustment that we document in the data.

The same mechanism also accounts for the other empirical facts. When volatility is high, revisions are driven more by learning about the inflation regime and less by the desire to remain close to the consensus, weakening the role of strategic alignment. Because forecasters receive heterogeneous private signals about the latent state, their posterior beliefs diverge more in volatile periods, generating greater cross-sectional disagreement. And although forecasts become more responsive, precision deteriorates because the underlying inflation process is both more volatile and harder to infer. In this sense, volatile times are characterized not only by more active updating, but also by noisier and less anchored expectations.

A central contribution of the paper is to discipline these mechanisms with micro data. We use moments from tranquil periods to identify the key structural parameters governing the three frictions: the fixed cost of revision, the strength of strategic concerns, and the informativeness of private signals. We then take the model to volatile episodes and ask how much of the observed change in micro moments it can account for once the economy is allowed to switch between low- and high-volatility regimes. This strategy lets the data discipline the frictions in normal times and then evaluates, out of sample, whether the interaction of those same frictions with regime-switching inflation can explain the behavior of forecasts in turbulent periods.

This empirical discipline also allows us to decompose the forces behind state-dependent forecasting. One part comes directly from higher inflation volatility, which makes the forecasting problem harder and raises forecast uncertainty. Another part comes from uncertainty about the regime itself: when forecasters must infer whether the economy has shifted into a different state, their beliefs become more dispersed and revision dynamics change even further. By comparing environments with and without regime uncertainty, we isolate the distinct roles of volatility and learning in generating responsiveness, disagreement, and forecast errors.

Contributions Our paper contributes to the literature on macroeconomic expectations by showing how aggregate inflation volatility reshapes the microeconomics of forecast updating.

First, we contribute to a growing literature showing that the response of economic decisions to shocks is state dependent, and in particular shaped by aggregate volatility. A central insight of [Vavra \(2014\)](#) is that higher uncertainty can make adjustment more frequent and thereby alter

the transmission of shocks. We show that a similar logic applies to professional inflation forecasts. Using survey microdata, we document that when inflation volatility is high, expectations become more responsive. In this sense, our paper brings the idea of state dependence from the adjustment literature to the formation of expectations, and shows that aggregate volatility changes not only the average level of expectations, but also their microeconomic dynamics.

More broadly, our evidence relates to a recent literature on state-dependent expectations. Pfäuti (2023) shows that when attention to inflation rises, inflation expectations become more sensitive to inflation, amplifying inflation persistence. Wang (2024) shows that agents update their inflation expectations more frequently as uncertainty rises. Using evidence on households' and firms' attention to the economy, Link, Peichl, Pfäuti, Roth and Wohlfart (2025) finds that more attentive respondents revise their inflation expectations more frequently after inflation shocks. For professional forecasters, Ahn and Farmer (2024) shows that disagreement at short horizons is driven by private information in normal times, while under high uncertainty it is driven more by heterogeneous responses to public information. Bianchi, Ilut and Saijo (2024) shows that professionals overreact more to acquired information in more uncertain times. We contribute to this literature by documenting a new set of state-dependent facts in survey microdata and organizing them through the lens of lumpy forecast adjustment.

Second, we contribute a micro-founded model of survey expectations that links aggregate volatility to individual forecast behavior. Relative to Baley and Turen (2024), the emphasis here is on how forecasting frictions interact with changing macroeconomic conditions over time. We build a Bayesian forecasting model with regime-switching inflation, fixed revision costs, strategic concerns, and private information about the latent state. We discipline the model using moments from tranquil periods to identify the key frictions, and then ask whether the same structure can account for the sharp changes in forecast behavior observed during volatile episodes. This provides a data-disciplined decomposition of the channels through which volatility matters, separating the direct effect of higher uncertainty from the additional effect of learning about regime shifts.

Third, our paper has implications for the interpretation of inflation surveys and for monetary policy. Survey forecasts are widely used by central banks as indicators of inflation persistence and anchoring, yet our results show that their meaning changes across states of the economy. More frequent revisions in volatile times need not imply that expectations are better informed or better anchored; they may instead reflect a combination of higher uncertainty, weaker consensus forces, and more active learning about the regime. This matters for how policymakers read shifts in survey expectations and for how they think about the transmission of inflation shocks. In that sense, our paper connects the literature on state-dependent adjustment and uncertainty (Vavra, 2014; Tenreyro and Thwaites, 2016) to the behavior of inflation expectations, offering a new perspective on how aggregate volatility shapes the propagation of shocks through beliefs.

2 Forecasting in Tranquil and Volatile Times

This section describes the inflation data, the definition of tranquil and volatile years, and the fixed-event forecasting framework. We then document five empirical facts showing that forecast dynamics are strongly state dependent.

2.1 Inflation

We measure annual inflation using the US Consumer Price Index (CPI). Let $cpi_{t,m}$ denote the CPI in month m of year t , and denote the yearly average with $\overline{cpi}_t \equiv \frac{1}{12} \sum_{m=1}^{12} cpi_{t,m}$. Then annual inflation in year t is

$$(1) \quad \pi_t = \frac{\overline{cpi}_t - \overline{cpi}_{t-1}}{\overline{cpi}_{t-1}} \approx \log(\overline{cpi}_t) - \log(\overline{cpi}_{t-1}).$$

Following [Giacomini, Skreta and Turen \(2020\)](#), annual inflation can also be decomposed into twelve monthly year-on-year contributions:

$$(2) \quad \pi_t \cong \sum_{m=1}^{12} x_{t,m}, \quad x_{t,m} \equiv \frac{\log(cpi_{t,m}) - \log(cpi_{t-1,m})}{12}, \quad m = 1, \dots, 12.$$

This decomposition is useful for our fixed-event setup because, as the year unfolds, part of annual inflation becomes realized while the rest remains to be forecasted. For some objects, especially rolling-window measures of inflation volatility, it is convenient to use a single monthly time index. Let τ denote calendar months in sequence, with a one-to-one mapping between τ and the pair (t, m) . Thus, $x_\tau \equiv x_{t,m}$ whenever month τ corresponds to month m of year t .

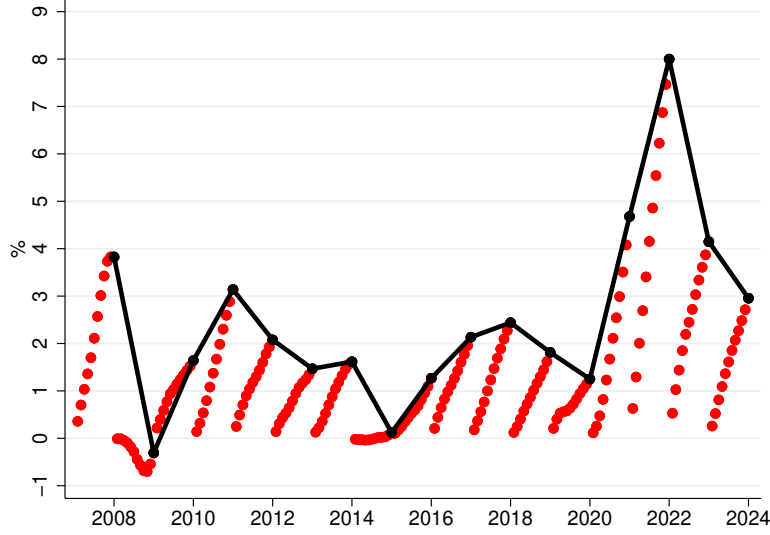
Figure [I](#) plots the series of annual inflation π_t (solid black line) and the partial sum of year-on-year monthly inflation $\sum_{m=1}^M x_{m,t}$ up to month M in year t (in red markers). By the relationship in [\(2\)](#), the two series coincide in the final month of each year ($M = 12$). In our sample, annual inflation averaged 2.45%, ranging from a low of -0.3% in 2009 to a high of 8.0% in 2022.

2.2 Two Inflation Regimes

Figure [II](#) summarizes the evolution of inflation over our sample. Panel (a) plots monthly year-on-year inflation, x_τ , while Panel (b) reports monthly inflation volatility, σ_τ^x , computed as the standard deviation of inflation over a 12-month rolling window:

$$(3) \quad \sigma_\tau^x \equiv \sqrt{\frac{1}{12} \sum_{j=0}^{11} (x_{\tau-j} - \bar{x}_\tau)^2}, \quad \bar{x}_\tau \equiv \frac{1}{12} \sum_{j=0}^{11} x_{\tau-j}.$$

Figure I – US Annual Inflation and Year-On-Year Monthly Inflation.



Notes: CPI inflation in the US for 2008–2024. Panel (a) plots the series of year-on-year monthly inflation $x_{m,t}$. Panel (b) plots the series of the annual inflation π_t (in a solid black line) and the partial sum of year-on-year monthly inflation $\sum_{m=1}^M x_{m,t}$ up to month M in year t (in red dots). Recall that for $M = 12$, the cumulated sum coincides with annual inflation, i.e., $\pi_t \cong \sum_{m=1}^{12} x_{m,t}$.

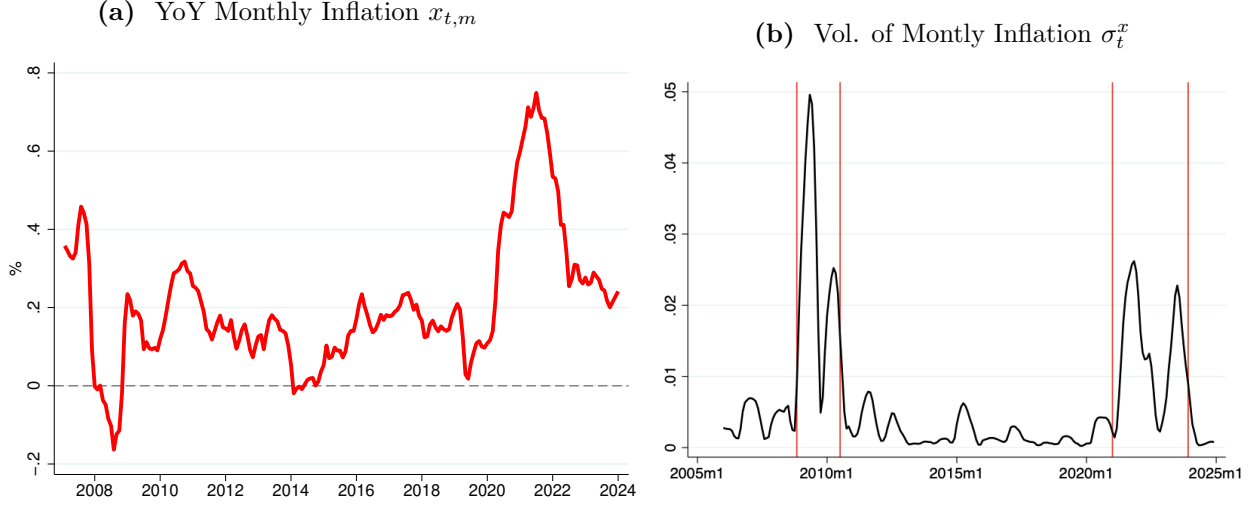
Motivated by these dynamics, we split the sample into *tranquil* years (2010–2020 and 2024) and *volatile* years (2008–2009 and 2021–2023). The volatile episodes correspond to the Great Recession and the post-pandemic inflation surge, two periods in which the inflation process was unusually unstable. Although both episodes are characterized by elevated volatility, they differ markedly in inflation levels: the first is associated with disinflationary and deflationary pressures, whereas the second is associated with persistently high inflation.

Table I summarizes the two regimes. Average inflation is 1.72 percent in tranquil years and 4.06 percent in volatile years, while inflation volatility doubles from 0.06 to 0.12. This empirical split motivates the two-state structure introduced in the model.

Table I – Two Inflation Regimes

		All years	Tranquil	Volatile
Periods		2008–2024	2010–2020, 2024	2008–2009, 2021–2023
Average Inflation	$\mathbb{E}[\pi]$	2.45	1.72	4.06
Inflation Volatility	$\sigma(\pi)$	0.07	0.06	0.12
Observations	N	17,663	9,256	8,407

Figure II – Inflation in the US



Notes: Bloomberg data. Normal years = 2010-2019 and 2024. Volatile years = 2008-2009 and 2021-23. Panel (a) is the rolling mean of the monthly inflation series x_h , and Panel (b) is the standard deviation. Shaded areas represent periods with heightened volatility.

2.3 Professional Forecasts

We analyze revisions of US annual CPI inflation forecasts from Bloomberg’s *Economic Forecasts ECFC* survey of professional forecasters. The survey includes participants from financial institutions and banks, consulting firms, universities, and research centers. A key advantage of the Bloomberg data is that users observe, in real time, each forecaster’s latest prediction, the date of the last update, and the consensus forecast.

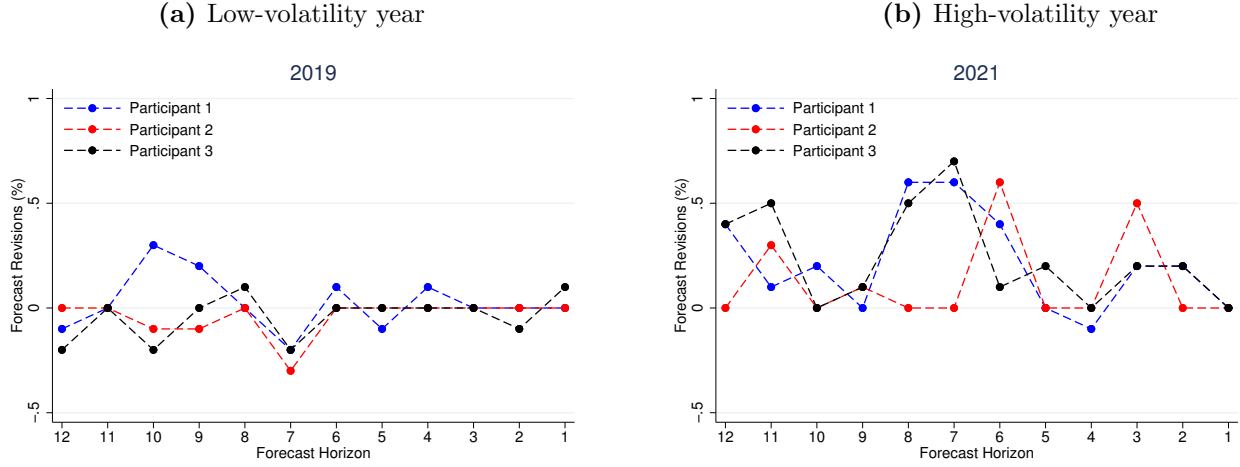
Fixed-event forecasts. We focus on monthly fixed-event forecasts of annual US inflation. For each year, we retain participants who report forecasts for the 12 months leading up to the release of the final annual inflation figure, and we exclude forecasters with no annual inflation revision. This leaves roughly 100 forecasters per year. The panel contains the history of forecast updates over the 18-month window preceding the end of each year.

Within a given year t , let f_h^i , for $i = 1, \dots, N$ and $h = 12, \dots, 1$, denote forecaster i ’s prediction made h months before the end of the year. Horizons are indexed backward, so that larger h corresponds to earlier forecasts. At horizon h , the forecast can be written as

$$(4) \quad f_h^i = \underbrace{\mathcal{P}_h^i}_{\text{projection of remaining inflation}} + \underbrace{\sum_{j=h+1}^{12} x_j}_{\text{realized inflation}} .$$

Thus, a fixed-event forecast combines a projection for the unresolved component of annual inflation with the observed realizations.

Figure III – Three forecasters in low and high volatility years



Notes: The figure shows forecast revisions for three representative professional forecasters across horizons in a low-volatility year (2019) and a high-volatility year (2021), in percentage points. Revisions are small, infrequent and correlated in 2019, but larger, more frequent, and dispersed in 2021.

Figure III provides a simple visual illustration of the main mechanism in the data. It plots the sequence of monthly forecast revisions for three forecasters in one tranquil year, 2019, and one volatile year, 2021.

In the tranquil year shown in panel (a), revisions are generally small, and many months exhibit no change. Forecast paths are relatively smooth, with long stretches of inaction interrupted by occasional modest adjustments. In the volatile year shown in panel (b), the picture is very different. Revisions are more frequent, their magnitude is larger, and the three forecasters move less synchronously. The same forecasters who were relatively inactive in 2019 adjust their forecasts repeatedly in 2021. Thus, volatile periods are associated not only with greater updating but also with greater heterogeneity in individual responses. The subsequent tables and regression-based results show that these differences are systematic in the full sample.

2.4 Forecast Revisions

For a given year, the revision at horizon h is the change in the forecast between two consecutive horizons: $\Delta f_h^i \equiv f_h^i - f_{h+1}^i$.

Table II summarizes the distribution of forecast revisions across tranquil and volatile periods. Several patterns stand out. First, revisions are substantially larger in volatile years. Conditional on revising, the average absolute change in the forecast rises from 0.247 in tranquil periods to 0.453 in volatile periods, an increase of more than 80%. Second, revisions are much more dispersed: the variance of revisions increases from 0.055 to 0.227, indicating that volatile periods are characterized not only by more adjustment, but also by greater heterogeneity. Third, forecasts become more active along the extensive margin. The average number of revisions per year rises from 5.06 in tranquil periods to 5.6 in volatile periods, while the inaction rate falls sharply from 56.9% to

Table II – Summary Statistics of Forecast Revisions

		All	Tranquil	Volatile
Average	$\mathbb{E}(\Delta f)$	−0.002	−0.013	0.028
Size	$\mathbb{E}(\text{abs}(\Delta f) \Delta f \neq 0)$	0.307	0.247	0.453
Dispersion	$\mathbb{V}ar(\Delta f)$	0.104	0.055	0.227
Number of revisions	$\text{count}(\Delta f \neq 0)$	5.204	5.059	5.602
Duration (months)	$\mathbb{E}(\tau)$	1.497	1.594	1.231
Inaction rate	$\Pr(\Delta f = 0)$	0.523	0.569	0.392
Observations	N	12,619	9,256	3,363

Notes: Bloomberg data. Tranquil = 2010-2020 and 2024. Volatile= 2008-09 and 2021-23. Averages across years and horizons.

39.2%. In addition, the average duration of inaction spells declines from 1.594 months in tranquil periods to 1.231 months in volatile periods. Finally, the mean revision itself changes sign across regimes: it is slightly negative in tranquil years (−0.013) and positive in volatile years (0.028). This pattern is consistent with the recent volatile episode being dominated by upward inflation surprises.

2.5 State-Dependent Forecasts

The summary statistics above indicate that, on average, forecast revisions are more active during volatile periods. We now examine how this state dependence appears in the time series and across horizons. Doing so is useful because a fixed-event forecasting environment has a natural within-year structure: as the horizon shortens, more inflation is realized, and less remains to be forecasted. We uncover five empirical facts. In volatile times, forecasts become less sticky: forecasters revise more often and by larger amounts, rely less on the consensus, and yet end up with lower precision and greater disagreement. These patterns motivate the state-dependent forecasting model developed in the next section.

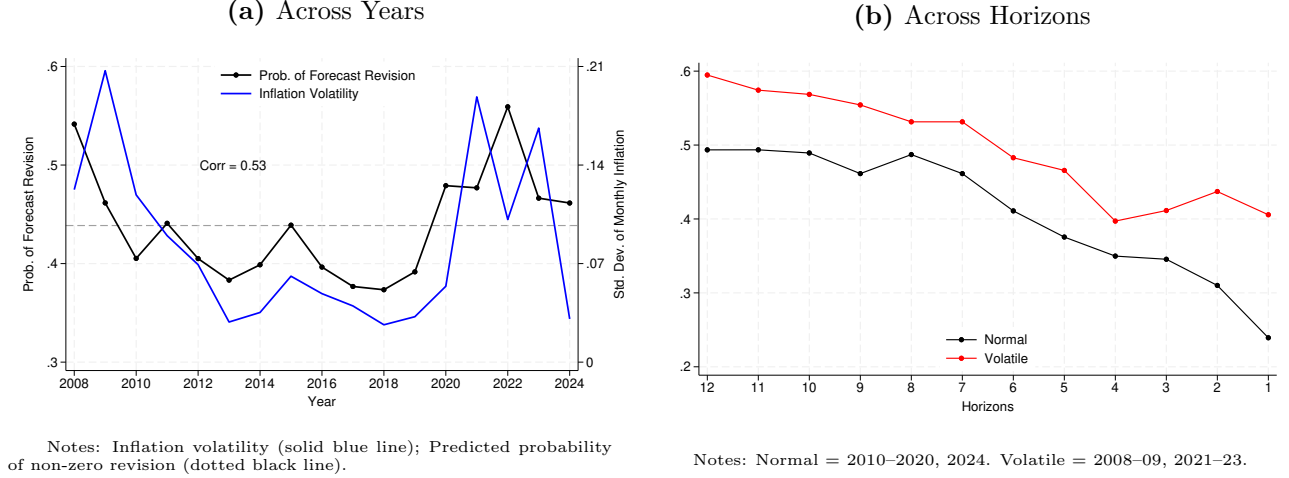
Fact I: More frequent revisions in volatile times. We begin with the extensive margin of adjustment. For each year, we estimate the probability of a non-zero forecast revision using the linear probability model

$$(5) \quad \Pr(\Delta f_{h,t}^i \neq 0) = c + \alpha_i + \alpha_h + \alpha_t + \epsilon_{iht},$$

where α_i , α_h , and α_t denote forecaster, horizon, and year fixed effects. We interpret $\hat{c} + \hat{\alpha}_t$ as the average predicted probability of revising in year t , conditional on the composition of forecasters and horizons. Figure IVa shows that the yearly revision probability co-moves strongly with inflation

volatility. The correlation between the two series is 0.53, and revision frequency peaks in the volatile years. Figure IVb shows the same pattern across horizons: forecasters revise more often in volatile years at every horizon.

Figure IV – Revision Frequency in Normal and Volatile Years

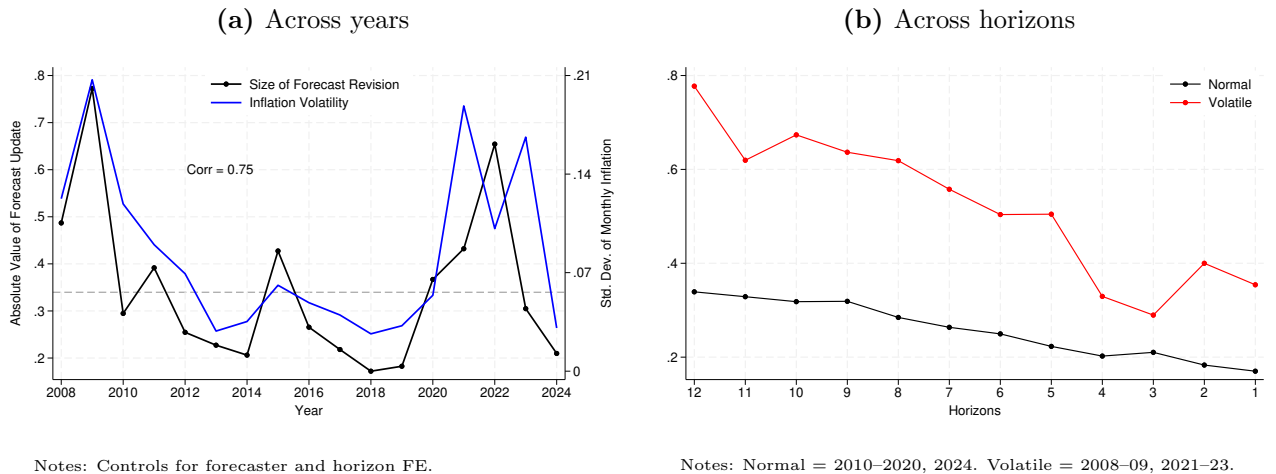


Fact II: Larger revisions in volatile times. We next turn to the intensive margin. Using the same specification as in (5), we replace the dependent variable with the absolute size of revisions, conditional on adjustment:

$$(6) \quad |\Delta f_{h,t}^i| = c + \alpha_i + \alpha_h + \alpha_t + \epsilon_{iht}, \quad \text{conditional on } \Delta f_{h,t}^i \neq 0.$$

Figure Va shows that revision size also co-moves strongly with inflation volatility across years, with a correlation of 0.75. Figure Vb shows that revisions are larger in volatile years at every horizon, especially at long horizons, when a larger fraction of annual inflation remains to be forecasted.

Figure V – Revision Size in Normal and Volatile Years



Fact III: Less alignment with the consensus in volatile times. Forecast revisions may also reflect strategic concerns. To study this force, we construct the consensus gap as the difference between forecaster i 's lagged forecast and the current consensus forecast F_h :

$$(7) \quad c_h^i \equiv f_{h+1}^i - F_h, \quad \text{where} \quad F_h \equiv \frac{1}{N} \sum_{i=1}^N f_h^i.$$

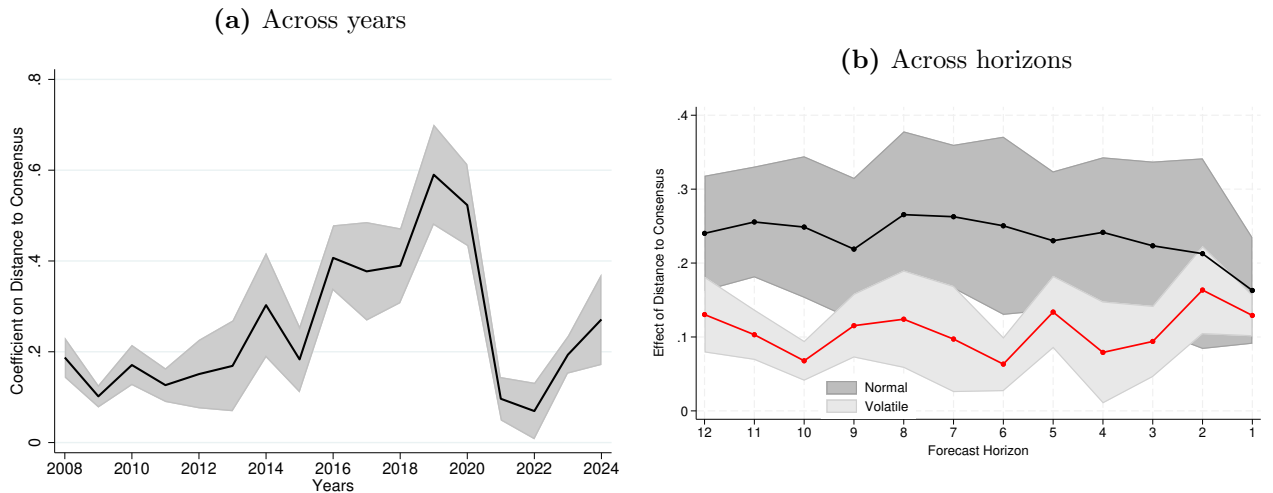
We then estimate

$$(8) \quad \Pr(\Delta f_{t,h}^i < 0) = \beta_0 + \beta_{h,v}(c_{t,h}^i \times \alpha_h \times \alpha_v) + \alpha_i + \alpha_t + \epsilon_{t,h}^i,$$

where α_v is a dummy for volatile years. Thus, we focus on the probability of a *downward* revision as a function of the forecaster's distance to the consensus. The corresponding results for upward revisions are very similar, indicating that the main pattern is not driven by asymmetries in the direction of adjustment.

Figure VIa shows that the effect of distance to consensus is much stronger in tranquil years. From roughly 2010 to 2020, the coefficient on the consensus gap rises and remains clearly positive, indicating that forecasters farther from the consensus are more likely to revise downward. This pattern collapses in 2021–23, returning to values similar to those observed in 2008–09. Figure VIb shows the same result across horizons: the sensitivity of revision decisions to the consensus gap is systematically weaker in volatile periods, especially at medium and long horizons. Quantitatively, a one-percentage-point larger distance from the consensus raises the probability of a downward revision by about 30% in tranquil years, but by only about half as much in volatile years. The corresponding estimates for upward revisions are similar in magnitude and lead to the same conclusion.

Figure VI – Effect of Consensus Gap on Probability of Downward Revision



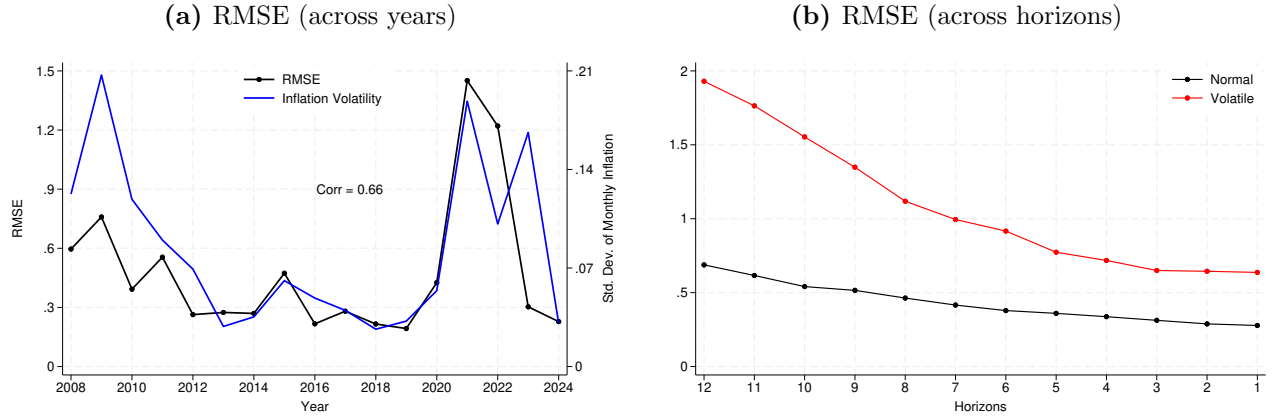
Notes: Controls for forecaster and horizon FE.

Fact IV: Lower precision in volatile times. Forecast accuracy deteriorates sharply in volatile periods. Let

$$(9) \quad e_h^i \equiv \pi - f_h^i$$

denote the ex-post forecast error of forecaster i at horizon h . We summarize precision using the root mean squared error (RMSE). The RMSE rises from 0.43 in tranquil periods to 1.09 in volatile periods, an increase of about 92%. Figure VIIa shows that RMSE co-moves strongly with inflation volatility across years, with a correlation of 0.66. Figure XVIb shows that this deterioration in precision is present at all horizons and is especially pronounced at long horizons. As the year progresses and more inflation is realized, RMSE declines in both regimes, but it remains substantially higher in volatile years throughout.

Figure VII – Forecast Precision in Normal and Volatile Years



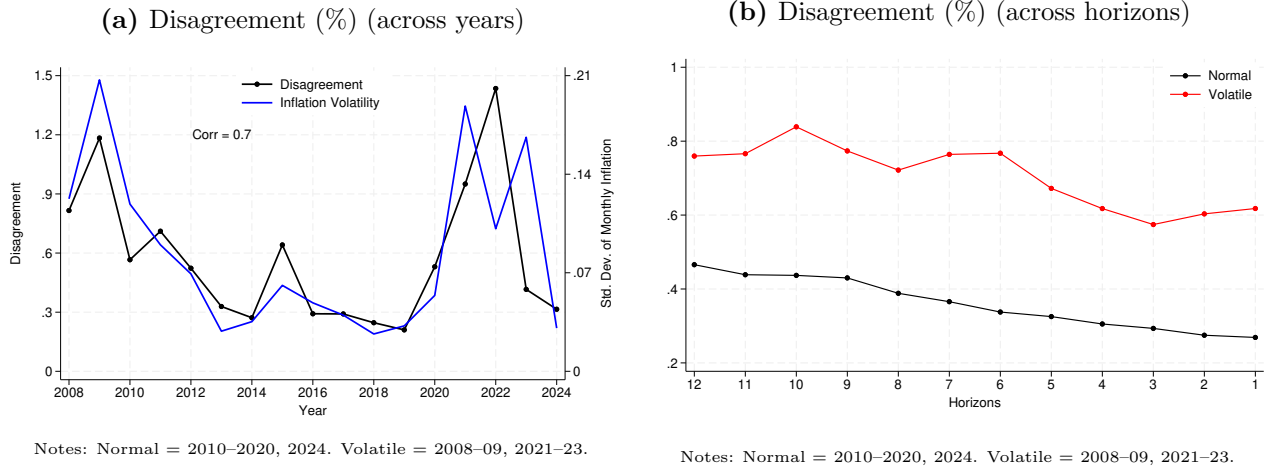
Notes: Controls for forecaster and horizon FE.

Notes: Normal = 2010–2020, 2024. Volatile = 2008–09, 2021–23.

Fact V: More disagreement in volatile times. Volatile periods are also characterized by substantially more cross-sectional disagreement, defined as the cross-sectional standard deviation of forecasts at horizon h , $Std(f_h^i)$. Average disagreement rises from 0.36 in tranquil periods to 0.70 in volatile periods, roughly doubling. Figure VIIIa shows that disagreement co-moves strongly with inflation volatility across years, with a correlation of 0.70. Figure VIIIb shows that disagreement is higher in volatile periods at every horizon. Volatile times therefore not only reduce average forecast precision, but also increase heterogeneity in beliefs across forecasters.

Summing up. The five facts point to a clear conclusion. When inflation becomes volatile, professional forecasts become less lumpy and less anchored to the consensus, but also less precise and more dispersed. Any model aimed at matching the recent inflation episode must therefore account for both margins: greater responsiveness and lower precision.

Figure VIII – Forecast Disagreement in Normal and Volatile Years



3 A Model of State-Dependent Forecasts

We develop a fixed-event Bayesian forecasting model with private information, frequent information revelation, fixed revision costs, and strategic concerns. Forecasters predict end-of-year inflation, while current monthly inflation is only imperfectly inferred and the inflation process switches between two latent regimes. The key ingredients are: (i) a two-state Markov-switching AR(1) process for monthly inflation, (ii) private signals about the latent regime, and (iii) lumpy forecast updating due to a fixed revision cost.

3.1 Baseline Forecasting Problem

A continuum of forecasters, indexed by $i \in N$, report forecasts of annual inflation π . Time within a calendar year is indexed by horizons $h \in \{1, \dots, 12\}$, where $h = 12$ denotes the first month of the year and $h = 1$ the last.

Payoffs. Let f_h^i denote forecaster i 's forecast of end-of-year inflation at horizon h . Changing a forecast entails paying a fixed revision cost $\kappa > 0$, measured in utility units. For a given initial forecast f_{13}^i , forecasters solve

$$(10) \quad \min_{\{f_h^i\}_{h=12}} \mathbb{E} \left[\underbrace{\sum_{h=12}^1 (f_h^i - \pi)^2}_{\text{accuracy}} + \underbrace{r (f_h^i - F_h)^2}_{\text{strategic}} + \underbrace{\kappa \mathbb{1}_{\{f_h^i \neq f_{h+1}^i\}}}_{\text{stability}} \middle| \mathcal{I}_0^i \right].$$

The first term captures losses from forecast errors. The second captures strategic concerns relative to the consensus forecast F_h : if $r > 0$, forecasters prefer to stay close to the consensus; if $r < 0$, they prefer to deviate from it. The third term is a fixed cost of revision that induces inaction. We interpret κ and r as structural parameters that are constant across aggregate states.

3.2 Recasting in Projection Terms

Annual inflation is the sum of within-year monthly inflation rates: $\pi \equiv \sum_{h=1}^{12} x_h$. At horizon h , this object can be decomposed into a realized, or sunk, component that is already known and a remaining component yet to be observed:

$$(11) \quad \pi = \underbrace{\sum_{k=h+1}^{12} x_k}_{\mathcal{S}_h} + \underbrace{\left(x_h + \sum_{k=1}^{h-1} x_k \right)}_{\text{remaining inflation}}.$$

The first term, $\mathcal{S}_h \equiv \sum_{k=h+1}^{12} x_k$, is common knowledge at horizon h . Forecaster i 's belief about the unresolved component is the projection

$$(12) \quad \mathcal{P}_h^i \equiv \mathbb{E} \left[x_h + \sum_{k=1}^{h-1} x_k \middle| \mathcal{I}_h^i \right].$$

Hence expected annual inflation can be written as

$$(13) \quad \mathbb{E}[\pi | \mathcal{I}_h^i] = \mathcal{S}_h + \mathcal{P}_h^i.$$

Since $\mathcal{S}_h$ is already known, it is convenient to net it out from both the individual forecast and the perceived consensus forecast:

$$(14) \quad \tilde{f}_h^i \equiv f_h^i - \mathcal{S}_h, \quad \tilde{F}_h \equiv F_h - \mathcal{S}_h.$$

The forecasting problem can therefore be recast as choosing a projection for the unresolved component of annual inflation:

$$(15) \quad \min_{\{\tilde{f}_h^i\}_{h=12}^1} \mathbb{E} \left[\sum_{h=12}^1 (\tilde{f}_h^i - \mathcal{P}_h^i)^2 + r(\tilde{f}_h^i - \tilde{F}_h)^2 + \kappa \mathbb{1}_{\{\tilde{f}_h^i \neq \tilde{f}_{h+1}^i - x_{h+1}\}} \middle| \mathcal{I}_0^i \right].$$

The relevant adjustment margin is therefore whether the forecaster revises the remaining-year projection $\tilde{f}_h^i$ relative to the inherited projection $\tilde{f}_{h+1}^i - x_{h+1}$. This representation isolates the object that actually matters for dynamic decision-making: the forecast of inflation that is still unresolved at horizon h .

3.3 Latent Regimes, Information, and Private Signals

Monthly inflation follows a two-state Markov-switching autoregressive process:

$$(16) \quad x_{h-1} = c_{u_{h-1}} + \phi_{u_{h-1}} x_h + \sigma_{u_{h-1}} \varepsilon_{h-1}, \quad \varepsilon_{h-1} \sim \mathcal{N}(0, 1),$$

where the latent regime $u_h \in \{L, H\}$ evolves according to the Markov chain

$$(17) \quad P = \begin{pmatrix} p_{LL} & p_{LH} \\ p_{HL} & p_{HH} \end{pmatrix}, \quad p_{ij} = \Pr(u_{h-1} = j \mid u_h = i).$$

State L is a low-volatility regime and state H is a high-volatility regime. The regime affects the conditional mean, persistence, and innovation variance through $(c_{u_h}, \phi_{u_h}, \sigma_{u_h})$.

At horizon h , forecaster i observes the latest realized monthly inflation x_{h+1} , which is public data, and receives a private discrete signal $s_h^i \in \{0, 1\}$ about the current regime u_h . Thus, the forecaster's information set at horizon h is

$$(18) \quad \mathcal{I}_h^i \equiv \left\{ \mathcal{I}_h^{\text{pub}}, s_h^i \right\},$$

where $\mathcal{I}_h^{\text{pub}} \equiv \{x_{h+1}, x_{h+2}, \dots, F_{h+1}, F_{h+2}, \dots\}$ contains publicly observed variables up to horizon h , including the latest realized inflation x_{h+1} and past consensus information. The private signal satisfies

$$(19) \quad \Pr(s_h^i = 1 \mid u_h = H) = \ell_H, \quad \Pr(s_h^i = 0 \mid u_h = L) = \ell_L.$$

The parameters ℓ_H and ℓ_L measure the informativeness of the private signal about the latent regime. This private information is crucial for generating heterogeneous posterior beliefs and, therefore, heterogeneous revisions across forecasters, especially in volatile periods.

Given $\mathcal{I}_h^i$, forecaster i forms a posterior belief about the current regime:

$$(20) \quad \hat{p}_h^i \equiv \Pr(u_h = H \mid \mathcal{I}_h^i).$$

This posterior summarizes the forecaster's assessment that the economy is currently in the high-volatility regime. It is the key belief state governing both the forecast of future inflation and the decision to revise. We derive its Bayesian updating formulas explicitly in Section 3.6.1 below.

3.4 Timing and Forecast Projections

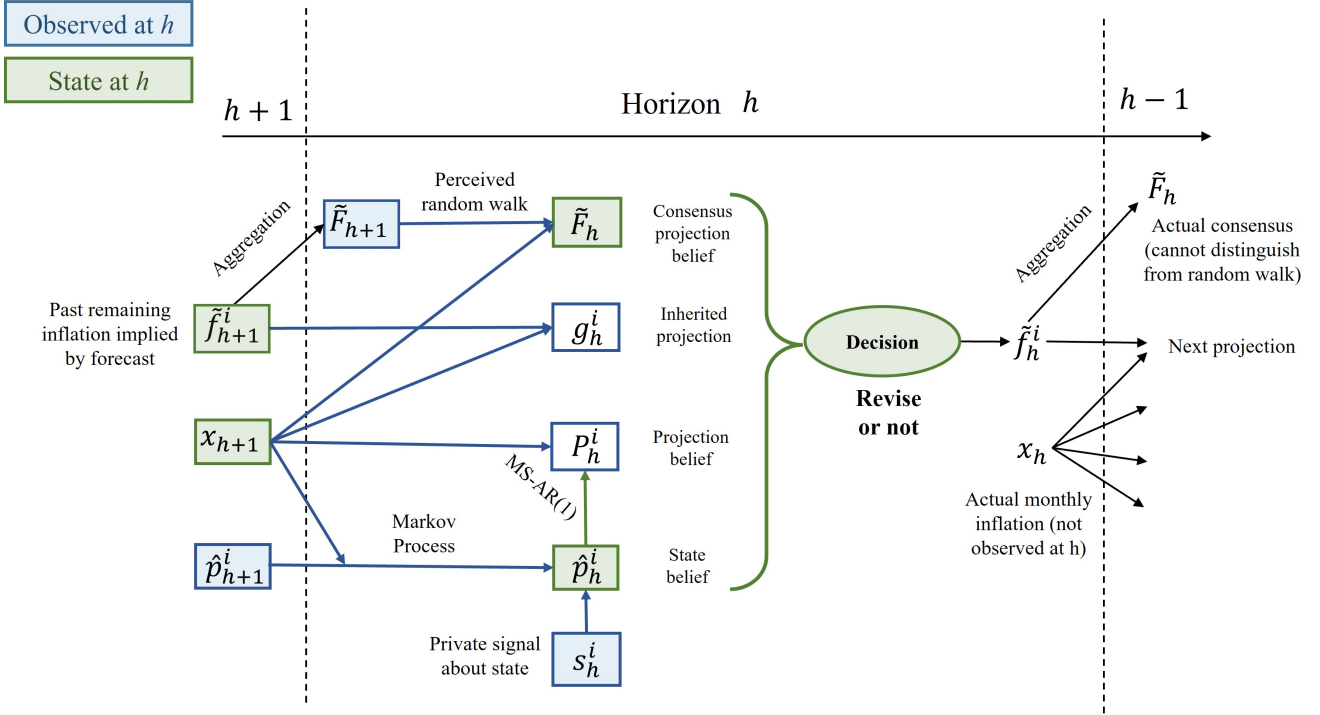
At horizon h , forecaster i enters the period with last period's chosen remaining-year projection $\tilde{f}_{h+1}^i$ and current information set $\mathcal{I}_h^i$. Once the newly released inflation realization x_{h+1} becomes known, the inherited remaining-year projection is

$$(21) \quad \tilde{f}_{h+1}^i - x_{h+1},$$

which is the projection that prevails if the forecaster does not update.

The timing within horizon h is as follows (see Figure IX):

Figure IX – Timing and Information Updating



Notes: The figure illustrates the timeline of information revelation, belief formation, and forecast revisions for three contiguous horizons $h+1$, h , and $h-1$.

1. The forecaster observes the new public realization x_{h+1} , which becomes part of the sunk component $\mathcal{S}_h$ and enters the public dataset.
2. The forecaster receives a private signal s_h^i about the current regime u_h .
3. The forecaster observes past consensus information F_{h+1} and forms the perceived consensus projection $\tilde{F}_h$.
4. Combining public x_{h+1} and the private s_h^i with the prior $\hat{p}_{h+1}^i$, the forecaster forms the posterior belief $\hat{p}_h^i$ in (20).
5. Given $\hat{p}_h^i$, the forecaster computes the remaining-year inflation projection $\mathcal{P}_h^i$ in (12).
6. The forecaster either keeps the inherited projection or chooses a new projection $\tilde{f}_h^i$.

This timing makes clear that the relevant state variable for adjustment is not the annual forecast itself, but the unresolved component of annual inflation. The sunk component $\mathcal{S}_h$ is common across forecasters, whereas heterogeneity arises from differences in inherited projections and in posterior beliefs. The latter reflect both heterogeneous private signals and the forecaster's inference about the latent regime. In the quantitative section below, we solve this Bayesian updating problem explicitly and show how it maps into forecast revisions.

3.5 Bellman Equation

Let V_h denote the value function at horizon h , with terminal condition $V_0(\cdot) = 0$. The individual state vector is

$$(22) \quad \left(\tilde{f}_{h+1}^i, \tilde{F}_h, \hat{p}_h^i, x_{h+1} \right),$$

where $\tilde{f}_{h+1}^i$ is the inherited remaining-year projection, $\tilde{F}_h$ is the perceived consensus projection, $\hat{p}_h^i$ is forecaster i 's posterior belief that the economy is in the high-volatility regime, and x_{h+1} is the latest realized monthly inflation.

The Bellman equation is

$$(23) \quad V_h(\tilde{f}_{h+1}^i, \tilde{F}_h, \hat{p}_h^i, x_{h+1}) = \min_{\tilde{f}_h^i} \left\{ (\tilde{f}_h^i - \mathcal{P}_h^i)^2 + r(\tilde{f}_h^i - \tilde{F}_h)^2 + \kappa \mathbb{I}_{\{\tilde{f}_h^i \neq \tilde{f}_{h+1}^i - x_{h+1}\}} \right. \\ \left. + \mathbb{E} \left[V_{h-1}(\tilde{f}_h^i, \tilde{F}_{h-1}, \hat{p}_{h-1}^i, x_h) \mid \mathcal{I}_h^i \right] \right\}.$$

The continuation value depends on three transition objects. First, the perceived consensus evolves according to

$$(24) \quad \tilde{F}_{h-1} = \tilde{F}_h + \eta_{h-1} - x_h,$$

where η_{h-1} is an innovation to the perceived consensus projection. The shock η_{h-1} captures innovations to the perceived consensus projection under Restricted Perception Equilibrium: $\mathbb{E}[\eta_{h-1} \mid \mathcal{I}_h^i] = 0$ and $\text{Var}[\eta_{h-1} \mid \mathcal{I}_h^i] = \sigma_F^2$.

Second, beliefs about the latent regime evolve according to the updating operator

$$(25) \quad \hat{p}_{h-1}^i = U_h(\hat{p}_h^i, x_{h+1}; s_{h-1}^i, x_h),$$

where $U_h(\cdot)$ summarizes the Bayesian updating problem that combines realized inflation, the Markov transition of the latent regime, and the next private signal, characterized in Section 3.6.1.

Third, current monthly inflation is drawn from the predictive distribution

$$(26) \quad x_h \mid \mathcal{I}_h^i \sim G^i(x_h \mid x_{h+1}, \hat{p}_h^i),$$

where $G^i(\cdot)$ is the forecaster's regime-dependent predictive distribution for monthly inflation, derived in Section 3.6.2.

Thus, the continuation value integrates over realized inflation, consensus innovations, and future private information. Section 3.6.3 shows how these beliefs imply the remaining-year projection $\mathcal{P}_h^i$.

Static optimum. Conditional on adjustment, the static optimum is

$$(27) \quad \tilde{f}_h^{i,*} = \frac{\mathcal{P}_h^i + r\tilde{F}_h}{1+r}.$$

Absent revision costs, forecasts would move smoothly with beliefs and the consensus target. With fixed adjustment costs, however, the forecaster updates only when the gain from moving away from the inherited projection exceeds both the direct cost of revision and the option value of waiting. This generates an endogenous inaction region.

3.6 Learning About Inflation and the Latent Regime

Forecast revisions are driven by learning about a latent inflation regime. Forecasters observe realized monthly inflation with a lag and receive private signals about the underlying state of the economy. As a result, they update beliefs in two dimensions: they revise their assessment of whether the economy is in a tranquil or volatile regime and, conditional on that assessment, they revise beliefs about current and future inflation. In our framework, forecasters do not respond only to the level of inflation; they also respond to what each inflation realization reveals about the persistence and volatility of the underlying process.

3.6.1 Latent State Beliefs

Recall that $\hat{p}_h^i = \Pr(u_h = H \mid \mathcal{I}_h^i)$ denotes the posterior probability of being in the high-volatility regime. Updating from $\hat{p}_h^i$ to $\hat{p}_{h-1}^i$ occurs in three steps.

First, the forecaster learns from realized inflation. Some inflation outcomes are more likely under the volatile regime than under the tranquil one. Second, the forecaster uses the Markov transition matrix to map this updated belief into a predictive belief about next period's regime. Third, the forecaster combines that predictive prior with a new private signal. Public data, therefore, moves all forecasters in the same direction, while private information generates heterogeneity in posterior beliefs.

Step 1: Update with realized inflation. Once x_h is realized, the forecaster updates the belief about the current regime using Bayes' rule:

$$(28) \quad \begin{aligned} \bar{p}_h^i(\hat{p}_h^i, x_{h+1}; x_h) &= \Pr(u_h = H \mid \mathcal{I}_h^i, x_h) \\ &= \frac{\hat{p}_h^i \Pr(x_h \mid x_{h+1}, u_h = H)}{\hat{p}_h^i \Pr(x_h \mid x_{h+1}, u_h = H) + (1 - \hat{p}_h^i) \Pr(x_h \mid x_{h+1}, u_h = L)}. \end{aligned}$$

Inflation realizations therefore act as public signals about the latent regime. Unusually large or volatile realizations are relatively more likely under the high-volatility regime and push the

posterior toward that state. Benign realizations do the opposite. The strength of the update depends on how distinct the two regime-specific likelihoods are.

Step 2: Project one step ahead in the Markov chain. The posterior belief about the current regime is then mapped into a predictive belief for horizon $h - 1$:

$$(29) \quad \bar{p}_{h-1}^i(\hat{p}_h^i, x_{h+1}; x_h) = \Pr(u_{h-1} = H \mid \mathcal{I}_h^i, x_h) = (1 - \bar{p}_h^i)p_{LH} + \bar{p}_h^i p_{HH}.$$

This step captures persistence in the macroeconomic environment. If the volatile regime is persistent, then a high posterior probability of being in that regime today implies a high predictive probability of remaining there tomorrow. This persistence matters because it affects the expected future path of inflation and, therefore, the value of revising the end-of-year forecast today.

Step 3: Update with the new private signal. Finally, the forecaster receives a private signal s_{h-1}^i about the regime at horizon $h - 1$ and combines it with the predictive prior:

$$(30) \quad \begin{aligned} \hat{p}_{h-1}^i(\hat{p}_h^i, x_{h+1}; s_{h-1}^i, x_h) &= \Pr(u_{h-1} = H \mid \mathcal{I}_h^i, x_h, s_{h-1}^i) \\ &= \frac{\Pr(s_{h-1}^i \mid u_{h-1} = H) \bar{p}_{h-1}^i}{\Pr(s_{h-1}^i \mid u_{h-1} = H) \bar{p}_{h-1}^i + \Pr(s_{h-1}^i \mid u_{h-1} = L) (1 - \bar{p}_{h-1}^i)}. \end{aligned}$$

Because forecasters receive different private signals, they need not hold the same posterior even after observing the same inflation realization. This channel is central for generating the rise in disagreement observed in volatile periods.

Equations (29)–(31) define the updating operator $U_h(\cdot)$ in (25). Taken together, these steps imply that regime beliefs are nonlinear and state dependent: when inflation is more informative about the regime, or when the economy is more likely to be volatile, posterior beliefs move more sharply.

3.6.2 Current Monthly Inflation Beliefs

Given the posterior belief about the regime, forecaster i forms a predictive distribution for current monthly inflation. Because the inflation process is regime dependent, this object is a mixture of the regime-specific conditional laws of motion:

$$(31) \quad x_h \mid \mathcal{I}_h^i \sim G^i(x_h \mid x_{h+1}, \hat{p}_h^i) = \begin{cases} \mathcal{N}(c_L + \phi_L x_{h+1}, \sigma_L^2), & \text{with prob. } 1 - \hat{p}_h^i, \\ \mathcal{N}(c_H + \phi_H x_{h+1}, \sigma_H^2), & \text{with prob. } \hat{p}_h^i. \end{cases}$$

This predictive distribution summarizes the forecaster's beliefs about current inflation before the realization arrives. In tranquil periods, most of the probability mass is assigned to the low-volatility

component, so beliefs about inflation are relatively concentrated. In volatile periods, more mass is assigned to the high-volatility component, making the predictive distribution more dispersed and shifting expected inflation whenever the two regimes differ in conditional means.

3.6.3 Remaining-year projection

To construct the remaining-year projection $\mathcal{P}_h^i$, forecaster i iterates the regime-dependent law of motion for monthly inflation forward from horizon h . The key objects are the joint first moments of inflation and the latent regime.

At horizon h , define

$$(32) \quad m_h^{i,L} \equiv \mathbb{E}[x_h \mathbb{1}_{\{u_h=L\}} \mid \mathcal{I}_h^i], \quad m_h^{i,H} \equiv \mathbb{E}[x_h \mathbb{1}_{\{u_h=H\}} \mid \mathcal{I}_h^i].$$

The conditional mean of monthly inflation is then

$$(33) \quad \mathbb{E}[x_h \mid \mathcal{I}_h^i] = m_h^{i,L} + m_h^{i,H}.$$

Using the predictive mixture in (31), the initial joint moments are

$$(34) \quad m_h^{i,L} = (1 - \hat{p}_h^i)(c_L + \phi_L x_{h+1}), \quad m_h^{i,H} = \hat{p}_h^i(c_H + \phi_H x_{h+1}).$$

For future horizons ($s \geq 1$), beliefs about the latent regime evolve according to

$$(35) \quad \hat{p}_{h-s}^i = (1 - \hat{p}_{h-s+1}^i)p_{LH} + \hat{p}_{h-s+1}^i p_{HH},$$

and the corresponding joint first moments satisfy the recursions

$$(36) \quad m_{h-s}^{i,L} = (1 - \hat{p}_{h-s}^i)c_L + \phi_L \left(p_{LL}m_{h-s+1}^{i,L} + p_{HL}m_{h-s+1}^{i,H} \right),$$

$$(37) \quad m_{h-s}^{i,H} = \hat{p}_{h-s}^i c_H + \phi_H \left(p_{LH}m_{h-s+1}^{i,L} + p_{HH}m_{h-s+1}^{i,H} \right).$$

Therefore, at any future horizon $h - s$,

$$(38) \quad \mathbb{E}[x_{h-s} \mid \mathcal{I}_h^i] = m_{h-s}^{i,L} + m_{h-s}^{i,H}.$$

The remaining-year inflation projection is obtained by summing these conditional means over the unresolved months:

$$(39) \quad \mathcal{P}_h^i = \sum_{s=0}^{h-1} \mathbb{E}[x_{h-s} \mid \mathcal{I}_h^i].$$

Hence, expected annual inflation is

$$(40) \quad \mathbb{E}[\pi \mid \mathcal{I}_h^i] = \mathcal{S}_h + \mathcal{P}_h^i = \mathcal{S}_h + \sum_{s=0}^{h-1} \left(m_{h-s}^{i,L} + m_{h-s}^{i,H} \right),$$

where $\mathcal{S}_h$ is the sunk component of annual inflation already realized by horizon h .

This recursion highlights the source of state dependence in the model. A change in the posterior regime probability $\hat{p}_h^i$ affects not only beliefs about current inflation, but the entire expected path of future monthly inflation. As a result, changes in beliefs about the latent state propagate into the remaining-year projection and, through it, into the optimal forecast revision.

3.7 Expectation over One-Step-Ahead Variables

The continuation value in (23) reflects the fact that today's forecast choice affects tomorrow's inherited forecast, but tomorrow's state is itself uncertain. From the perspective of horizon h , three objects are not yet known: current monthly inflation x_h , next period's innovation to the perceived consensus η_{h-1} , and the private signal s_{h-1}^i that the forecaster will receive about the latent regime. Evaluating the continuation value therefore requires integrating over all three sources of uncertainty.

Conditional on choosing $\tilde{f}_h^i$, the continuation term is

$$(41) \quad \mathbb{E} \left[V_{h-1} \left(\tilde{f}_h^i, \tilde{F}_{h-1}, \hat{p}_{h-1}^i, x_h \right) \mid \mathcal{I}_h^i, \tilde{f}_h^i \right] \\ = \int \sum_{s_{h-1} \in \{0,1\}} \left[\int V_{h-1} \left(\tilde{f}_h^i, \tilde{F}_h + \sigma_F \eta_{h-1} - x_h, \hat{p}_{h-1}^i(\hat{p}_h^i, x_{h+1}; x_h, s_{h-1}), x_h \right) \phi(\eta_{h-1}) d\eta_{h-1} \right] \\ \times \Pr(s_{h-1} \mid x_h, \mathcal{I}_h^i) dG^i(x_h \mid x_{h+1}, \hat{p}_h^i).$$

where $\phi(\eta_{h-1})$ denotes the standard normal density. This expression shows how today's information maps into tomorrow's forecasting environment. The realization of x_h affects the extent to which annual inflation remains unresolved and shifts beliefs about the latent regime. New private information further perturbs those beliefs, creating heterogeneity across forecasters. At the same time, shocks to the perceived consensus move the strategic target. The continuation value, therefore, depends on both the evolution of fundamentals and the evolution of beliefs.

A key implication is that the next private signal is not independent of current inflation. Since x_h is informative about the regime, it changes the predictive probability of receiving signal $s_{h-1} = 1$:

$$(42) \quad \Pr(s_{h-1} = 1 \mid x_h, \mathcal{I}_h^i) = \bar{\bar{p}}_{h-1}^i(x_h) \ell_H + [1 - \bar{\bar{p}}_{h-1}^i(x_h)] (1 - \ell_L),$$

$$(43) \quad \Pr(s_{h-1} = 0 \mid x_h, \mathcal{I}_h^i) = 1 - \Pr(s_{h-1} = 1 \mid x_h, \mathcal{I}_h^i).$$

Thus, a high inflation realization today not only raises the likelihood that the economy is in the

volatile regime, but also increases the likelihood that tomorrow's private signal will confirm that view. Realized inflation and private information, therefore, reinforce each other through Bayesian learning.

Taken together, the pair $(x_{h+1}, \hat{p}_h^i)$ induces a unique predictive distribution for $(x_h, \hat{p}_{h-1}^i)$. This predictive distribution is the key object for solving the dynamic problem, because it determines both the expected benefit of revising today and the option value of waiting for additional information tomorrow.

3.8 Equilibrium

To close the model, we specify how forecasters perceive the evolution of the consensus forecast. Solving for the exact law of motion of the actual consensus would require a fixed point between individual forecasts and their aggregation. To preserve tractability, we adopt a *restricted perceptions equilibrium* (RPE), following [Evans and Honkapohja \(1993\)](#) and [Marcet and Nicolini \(2003\)](#).

In an RPE, forecasters are internally rational: given their perceived law of motion for the consensus, they solve the forecasting problem optimally. However, instead of knowing the exact equilibrium evolution of the consensus, they use a parsimonious forecasting rule that is statistically consistent with the realized consensus process ex post.

Definition 1. *A restricted perceptions equilibrium (RPE) consists of:*

- (i) *a perceived law of motion for the consensus projection $\{\tilde{F}_h\}$, given by*

$$(44) \quad \tilde{F}_h = g(\tilde{F}_{h+1}; \delta) + \epsilon_h^F, \quad \epsilon_h^F \sim \mathcal{N}(0, \sigma_F^2),$$

where σ_F^2 denotes the perceived variance of innovations to the consensus projection;

- (ii) *posterior regime beliefs $\{\hat{p}_h^i\}$, inflation projections $\{\mathcal{P}_h^i\}$, and forecast policies $\{\tilde{f}_h^i\}$ for all forecasters i and horizons h ;*

- (iii) *an actual consensus projection given by*

$$(45) \quad \tilde{F}_h = \frac{1}{N} \sum_{i=1}^N \tilde{f}_h^i.$$

such that:

1. *given the belief-updating rules, the perceived consensus process (44), and the state vector $(\tilde{f}_{h+1}^i, \tilde{F}_h, \hat{p}_h^i, x_{h+1})$, each forecast policy $\tilde{f}_h^i$ solves the Bellman problem in (23);*

2. the perceived law of motion is statistically consistent with the realized consensus process, in the sense that the forecast errors

$$(46) \quad \epsilon_h^F \equiv \tilde{F}_h - g(\tilde{F}_{h+1}; \delta)$$

satisfy $\mathbb{E}[\epsilon_h^F] = 0$, $\text{Cov}(\epsilon_h^F, \epsilon_j^F) = 0 \quad \forall h \neq j$, and $\text{Var}(\epsilon_h^F) = \sigma_F^2$.

In a restricted perceptions equilibrium, the perceived law of motion for the consensus need not coincide exactly with the true aggregation of individual forecasts. Instead, forecasters use the coefficients (δ, σ_F) that best summarize the realized consensus process within the restricted class (44). This assumption removes the fixed point between individual decisions and aggregate expectations while preserving the model's strategic force.

3.9 Optimal Forecasting Policy

The recursive problem implies a simple state-dependent policy. At each horizon h , a forecaster compares the value of retaining the inherited projection against paying the fixed cost and resetting the forecast. The optimal rule comprises two components: an inaction region and a reset forecast.

For each horizon h , define the inaction region $\mathcal{R}_h$ as the set of states for which it is optimal not to revise:

$$(47) \quad \mathcal{R}_h \equiv \left\{ (\mathcal{P}_h^i, \tilde{F}_h, \tilde{f}_{h+1}^i - x_{h+1}) : V_h^I \geq V_h^A \right\},$$

where V_h^I is the value of inaction and V_h^A is the value of adjustment. If the inherited projection lies inside $\mathcal{R}_h$, the forecaster leaves the forecast unchanged. If it lies outside, the forecaster pays the fixed cost and resets the projection to the optimal target

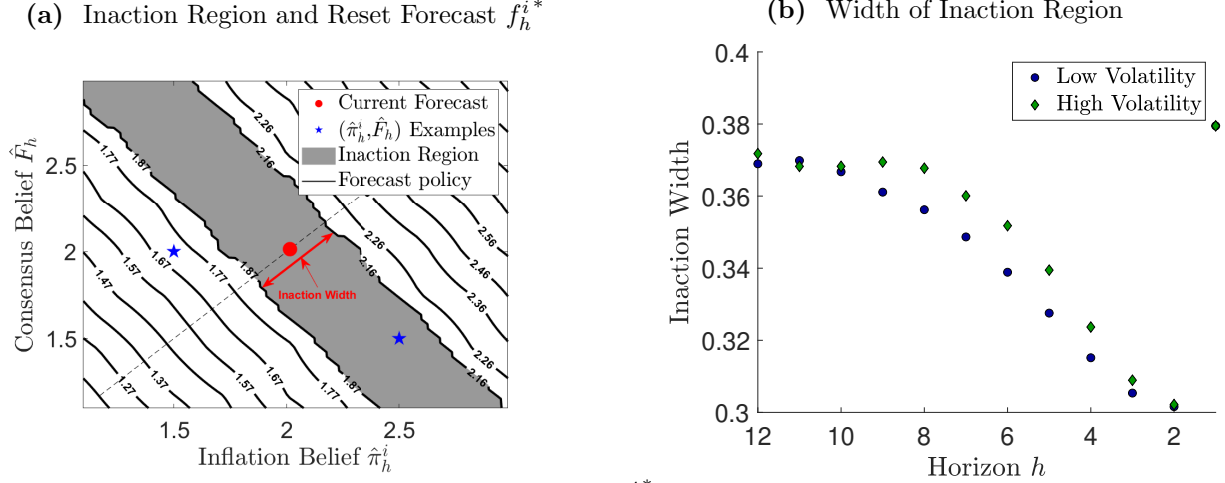
$$(48) \quad \tilde{f}_h^{i*} = \tilde{f}_h^{i*}(\mathcal{P}_h^i, \tilde{F}_h).$$

Hence, forecast revisions satisfy

$$(49) \quad \Delta \tilde{f}_h^i = \begin{cases} 0, & \text{if } \tilde{f}_{h+1}^i - x_{h+1} \in \mathcal{R}_h, \\ (\tilde{f}_h^i)^* - (\tilde{f}_{h+1}^i - x_{h+1}), & \text{otherwise.} \end{cases}$$

Panel (a) of Figure X illustrates the policy for a given horizon and inherited forecast. The shaded band is the inaction region: beliefs within the band leave the forecast unchanged, whereas beliefs outside the band trigger a reset. The contour lines show the reset forecast. The band slopes downward because beliefs about fundamentals and the consensus are partial substitutes. Panel (b) shows how its width varies across horizons and volatility regimes. At long horizons, uncertainty is greater and waiting is more valuable, which widens the inaction region through an *option effect*.

Figure X – Forecast Revision Policy: Inaction and Reset



Notes: The figure illustrates the reset forecast policy f_h^{i*} and inaction region $\mathcal{R}_h$ for $h = 6$ given a current forecast $f_h^i = 2$. We also show two examples of beliefs, one inside and one outside the inaction region. Panel (b) plots the inaction region width (the segment on the 45-degree line) for different horizons and two levels of inflation volatility σ_x , low and high.

Higher inflation volatility also makes beliefs move more across periods, increasing the probability of crossing the boundary. Volatile environments, therefore, combine a wider inaction region with more frequent boundary crossings. This interaction is central to the model, as it raises both the size and the frequency of revisions.

4 Quantitative Analysis

We now evaluate the model's quantitative performance. The strategy is to discipline the model using a small set of targeted moments, then assess whether it can replicate the broader empirical patterns observed in the data. This exercise serves two purposes. First, it allows us to pin down the key structural parameters governing forecast adjustment, strategic concerns, private information, and consensus uncertainty. Second, it provides a quantitative test of the mechanism: once calibrated to match a few central moments, can the model also reproduce the state-dependent facts on revisions, precision, and disagreement?

We proceed in two steps. We begin by calibrating the model, combining externally set parameters for the inflation process with internally calibrated behavioral parameters. We then study a set of untargeted moments, asking whether the model can match the empirical patterns across regimes and across horizons that were not directly imposed in the estimation.

4.1 Calibration

We split the parameterization into two blocks. First, we externally set the parameters governing the inflation process in each regime using reduced-form estimates from the data. Second, we

internally calibrate the behavioral parameters of the forecasting model to match key moments of forecast revisions and disagreement.

Externally set inflation-process parameters. The model is monthly. We discipline the inflation block using regime-specific estimates of the year-on-year monthly inflation process. Table III reports the parameterization of the monthly AR(1) process in tranquil and volatile periods, together with the implied annual moments.

During tranquil periods, the estimated monthly process is characterized by a mean of $c_x = 0.013$, a persistence of $\rho_x = 0.932$, and a volatility of $\sigma_x = 0.036$. In volatile periods, the inflation process becomes more persistent and substantially more volatile, with $\rho_x = 0.950$ and $\sigma_x = 0.054$. Although the monthly intercept is slightly lower, the higher persistence implies a higher annual mean inflation rate. Overall, the monthly process implies annual mean inflation of 2.23 percent in tranquil times and 2.64 percent in volatile times, annual persistence of 0.43 and 0.54, and annual volatility of 0.49 and 0.73, respectively.

Table III – Parameters of Two Inflation Regimes

		Monthly (x)		Yearly (π)		
		Tranquil	Volatile	Tranquil	Volatile	
Mean	c_x	0.013	0.011	c_π	2.23	2.64
Persistence	ρ_x	0.932	0.950	ρ_π	0.43	0.54
Volatility	σ_x	0.036	0.054	σ_π	0.49	0.73

Internally calibrated behavioral parameters. Conditional on the inflation process, we calibrate the remaining model parameters using simulated method-of-moments. These are the fixed revision cost κ , the strength of strategic concerns r , the informativeness of private regime signals (ℓ_L, ℓ_H) , and the perceived variance of innovations to the consensus projection, σ_F^2 .

The guiding idea is simple. The frequency and size of revisions discipline the degree of lumpiness in adjustment, strategic concerns discipline the extent to which forecasters respond to the consensus, and private signals help the model generate disagreement and state-dependent heterogeneity in revisions. We therefore choose parameters to match a set of moments that are informative about these margins. More specifically, we target the average probability of revision and the average size of revisions in tranquil periods, and two state-dependent moments: the ratio of forecast disagreement in volatile to tranquil periods, and the ratio of mean squared forecast errors across the two regimes.

Consensus consistency. Forecasters in the model do not know the exact equilibrium law of motion of the consensus forecast. Instead, under restricted perceptions equilibrium, they use a

Table IV – Internally calibrated parameters

Parameter	Value	Moment	Data		Model	
			Tranquil	Volatile	Tranquil	Volatile
κ	0.091	$\Pr[\Delta f \neq 0]$	0.43	0.50	0.43	0.55
r	0.119	$\mathbb{E}[\Delta f \mid \text{adj}]$	0.25	0.50	0.19	0.28
ℓ_L	0.383	$\text{Var}[f]$ ratio H/L		1.65		1.63
ℓ_H	0.903	MSFE ratio H/L		1.92		3.38

Notes: The table reports internally calibrated parameters and the corresponding moments in the data and the model. Tranquil = 2010–2020, 2024. Volatile = 2008–09, 2021–23. For moments reported as ratios H/L , the table shows the volatile-to-tranquil ratio in the data and in the model.

parsimonious forecasting rule for the consensus and treat innovations around that rule as Gaussian with variance σ_F^2 . We pin down σ_F^2 so that the perceived volatility of the consensus is internally consistent with the realized volatility generated by the simulated economy. Operationally, for a candidate value of σ_F^2 , we solve the individual forecasting problem by backward induction, simulate the model, compute the implied consensus process, and iterate until the perceived and realized consensus innovations are consistent.

Estimated parameters and model fit. Table IV reports the calibrated parameters and the corresponding model fit. The estimated fixed revision cost is $\kappa = 0.091$, confirming that forecast adjustment is subject to a quantitatively important friction. The estimate $r = 0.119$ implies positive but moderate strategic complementarities: forecasters care about staying close to the consensus, but this force is not dominant. The private-signal parameters, $\ell_L = 0.383$ and $\ell_H = 0.903$, imply that signals are informative because their distribution differs sharply across regimes.¹ Finally, the perceived variance of consensus innovations is $\sigma_F^2 = 0.054$.

The model provides a good fit to several key moments. It matches the average revision frequency in tranquil periods exactly, and comes close in volatile periods (0.55 in the model versus 0.50 in the data). It also closely matches the increase in disagreement across regimes, with a volatile-to-tranquil variance ratio of 1.63 in the model versus 1.65 in the data.

At the same time, the fit is not uniform across all moments. The model underpredicts the average size of revisions, especially in volatile periods (0.28 in the model versus 0.50 in the data), and it overpredicts the increase in mean squared forecast errors across regimes (3.38 in the model versus 1.92 in the data). We view these discrepancies as informative. They suggest that the model captures the direction and much of the magnitude of the state dependence in revisions and disagreement, but may still understate the force of large revisions in highly volatile episodes and overstate the deterioration in precision.

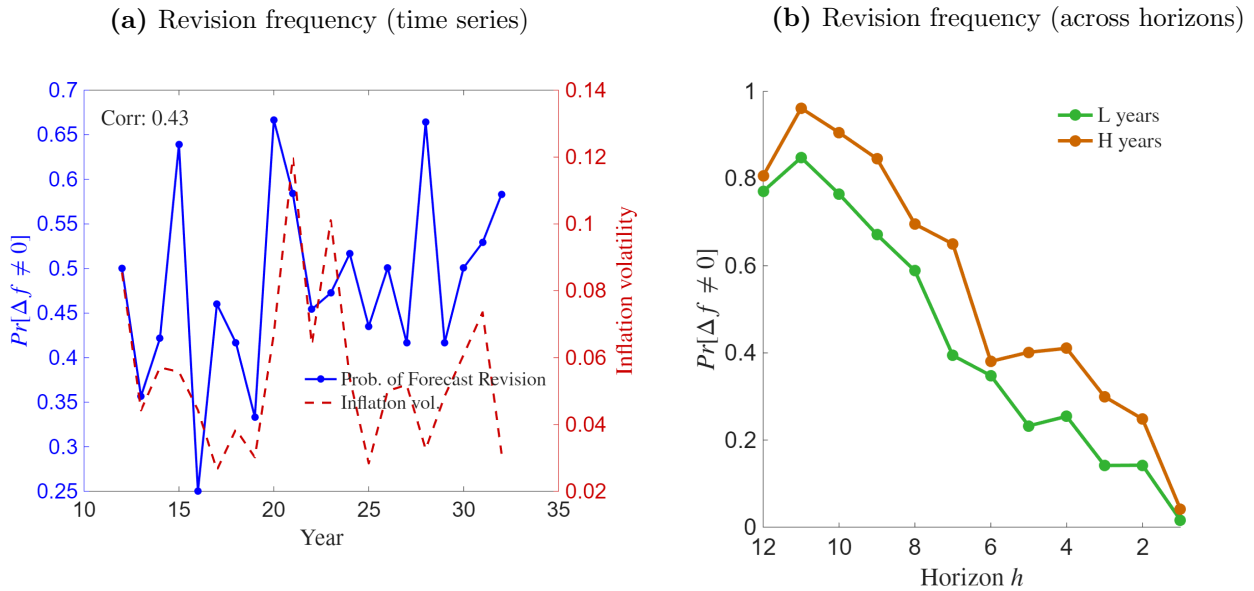
¹In particular, a high signal occurs with probability 0.903 in regime H , but only $1 - \ell_L = 0.617$ in regime L , making it about 46% more likely in the volatile regime. This separation helps the model generate stronger revisions and greater disagreement when volatility rises.

4.2 Results

We now ask whether the calibrated model can reproduce the state dependence in forecast behavior documented in the data. Figures XI–XV report the model-implied analogues of the five empirical facts in Figures IV–VIII. Recall that the model is intentionally disciplined on tranquil periods, so the volatile regime is the real test. Two questions are of interest. First, does the model generate the right qualitative shifts when the economy moves from low- to high-volatility periods? Second, how close does it come quantitatively to the magnitude of those shifts in the data?

Revision frequency. We start with the extensive margin. Figure IV in the data shows that revisions become more common in volatile times, both over time and across horizons; Figure XI shows that the model reproduces this pattern well. The time-series correlation with inflation volatility is positive, at 0.43, and the cross-horizon profile shifts up in the high-volatility regime at nearly every horizon. Quantitatively, the fit is quite good. The model matches the tranquil-period revision frequency essentially exactly and comes close in volatile times, where it predicts a revision probability of 0.55 compared with 0.50 in the data. So on the extensive margin, the model gets both the direction and much of the magnitude right.

Figure XI – Model-Generated Revision Frequency

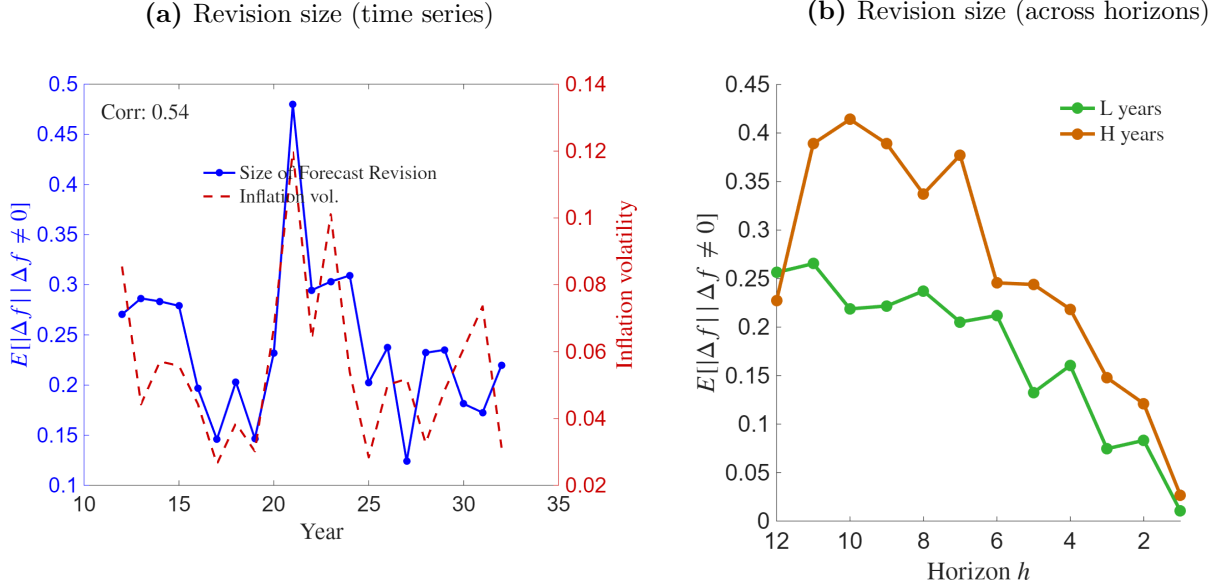


Notes: Model-generated revision frequency across years and forecast horizons for two inflation volatility regimes.

Revision size. Regarding the intensive margin, Figure XII shows that the model captures the broad pattern from Figure V: revisions are larger when volatility is high, and the gap is most visible at long horizons, where more of the year's inflation remains to be learned. The time-series

correlation with inflation volatility is also positive, at 0.54. But here the model is too muted. It generates the right ranking across regimes and horizons, yet it understates the size of revisions, especially in volatile periods. In the calibration moments, average revision size in the volatile regime is 0.28 in the model, against 0.50 in the data. This suggests that the model captures why volatile times trigger more adjustment, but quantitatively misses part of the force behind the largest revisions observed in the data.

Figure XII – Model-Generated Revision Size



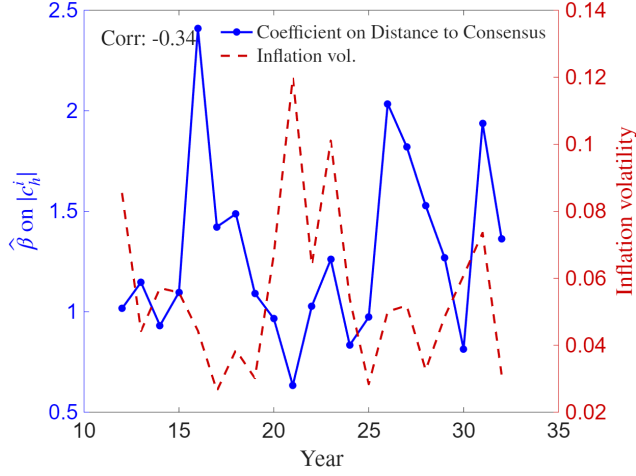
Notes: Model-generated revision size across years and forecast horizons for two inflation volatility regimes.

Strategic effect of the consensus gap. The model also reproduces the weakening role of the consensus in volatile times. In the data, Figure VI shows that distance to the consensus matters less for revision decisions when inflation is more volatile; the same pattern emerges in Figure XIII. Over time, the model-generated strategic coefficient is negatively correlated with inflation volatility, with a correlation of -0.34 , and across horizons the effect is systematically smaller in the high-volatility regime. When the inflation environment becomes more uncertain, revisions are driven less by strategic alignment and more by learning about the underlying state. The mechanism clearly goes in the right direction, even if the exact level of the strategic coefficient should not be overinterpreted one-for-one.

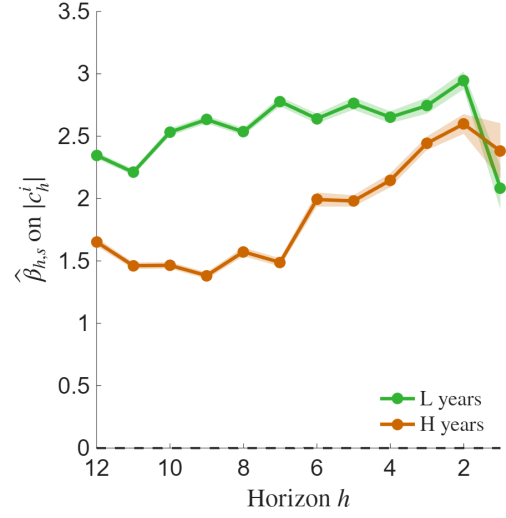
Forecast precision. Figure XIV shows that the model delivers larger forecast errors in volatile periods, just as in the data in Figure VII and the cross-horizon pattern is especially sharp at long horizons. The time-series correlation between RMSE and inflation volatility is strong, at 0.70. In that sense, the model captures the central tension in the data: forecasts become more responsive

Figure XIII – Model-Generated Strategic Effect of the Consensus Gap

(a) Effect of distance to consensus (time series)



(b) Effect of distance to consensus (across horizons)

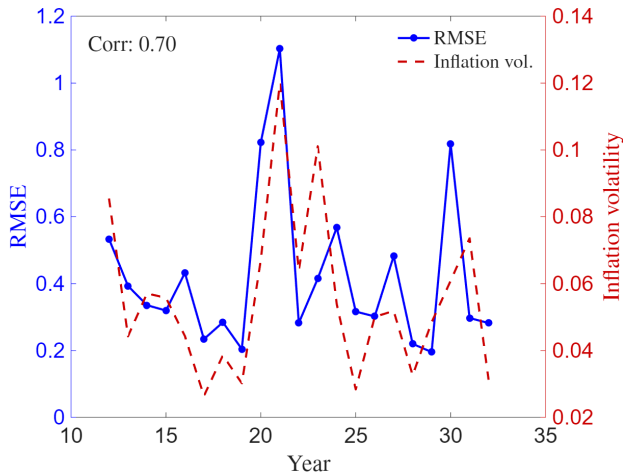


Notes: Model-generated effect of the consensus gap across years and forecast horizons for two inflation volatility regimes.

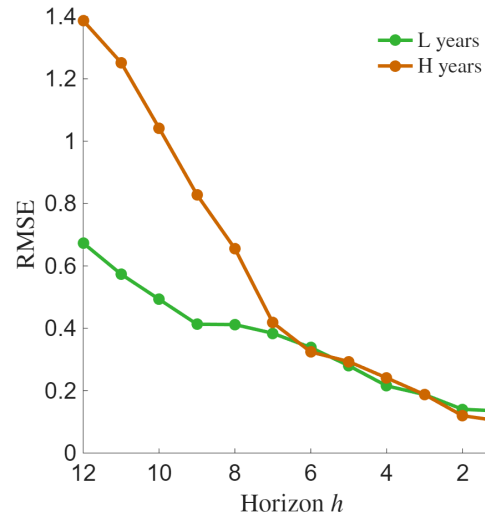
in turbulent times, but also less precise. Quantitatively, however, the model appears to overshoot the deterioration in precision. The increase in mean squared forecast errors across regimes is 3.38 in the model, compared with 1.92 in the data. So while the model gets the sign and the broad pattern right, it may attribute too much of the volatile-period dynamics to forecast uncertainty.

Figure XIV – Model-Generated Forecast Precision

(a) RMSE (time series)



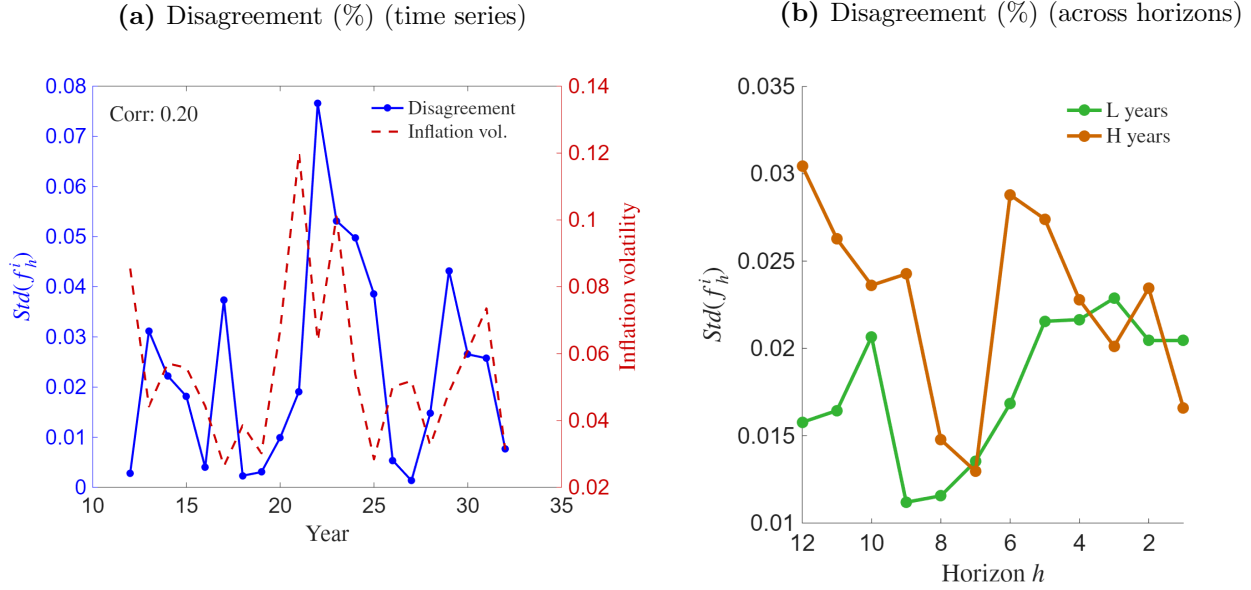
(b) RMSE (across horizons)



Notes: Model-generated RMSE across years and forecast horizons for two inflation volatility regimes.

Disagreement. Finally, the model performs well on disagreement. Figure XV shows higher dispersion of forecasts in the high-volatility regime, especially at medium and long horizons, in line with Figure VIII. The time-series correlation with inflation volatility is more modest, at 0.20, but the cross-regime comparison is quite successful quantitatively. In particular, the model closely matches the rise in disagreement across regimes: the volatile-to-tranquil variance ratio is 1.63 in the model and 1.65 in the data. This is an important success of the framework, since disagreement is a central implication of combining regime uncertainty with heterogeneous private signals.

Figure XV – Model-Generated Forecast Disagreement



Notes: Model-generated disagreement across years and forecast horizons for two inflation volatility regimes.

Taken together, these results paint a nuanced picture. The model does well in reproducing the shift in the frequency of revisions, the weakening role of the consensus, and the increase in disagreement, and it also captures the broad deterioration in forecast precision. Where it falls short is on the magnitude of revision size, which remains too small, and on forecast errors, where the model overshoots the rise observed in the data. Still, the overall message is clear: once disciplined on tranquil periods, the model goes a long way toward explaining why volatile times bring more active, less consensus-driven, less precise, and more dispersed forecast behavior.

In the next subsection, we use the model to decompose these patterns into the roles of higher volatility itself and uncertainty about regime changes.

5 Decomposing the Sources of State Dependence (In progress)

This section uses the model to decompose why forecast behavior changes in volatile times. In the data, volatile periods are characterized by more frequent and larger revisions, lower precision, and greater disagreement. In the model, these patterns can arise through two related forces. First, higher volatility directly makes inflation harder to predict, raising uncertainty and forecast errors. Second, higher volatility changes the value of adjustment in the presence of fixed revision costs and strategic concerns, altering the optimal timing and size of revisions.

Our baseline specification combines both channels through learning about the latent regime. Forecasters do not observe regime changes directly. Instead, they use realized inflation and private signals to infer whether the economy is in the tranquil or volatile state. As a result, forecast behavior adjusts gradually rather than discretely: when volatility rises, agents initially remain uncertain about the new environment, and only over time do they update their beliefs and revise their policies accordingly. This baseline provides the natural benchmark for interpreting state-dependent forecast dynamics.

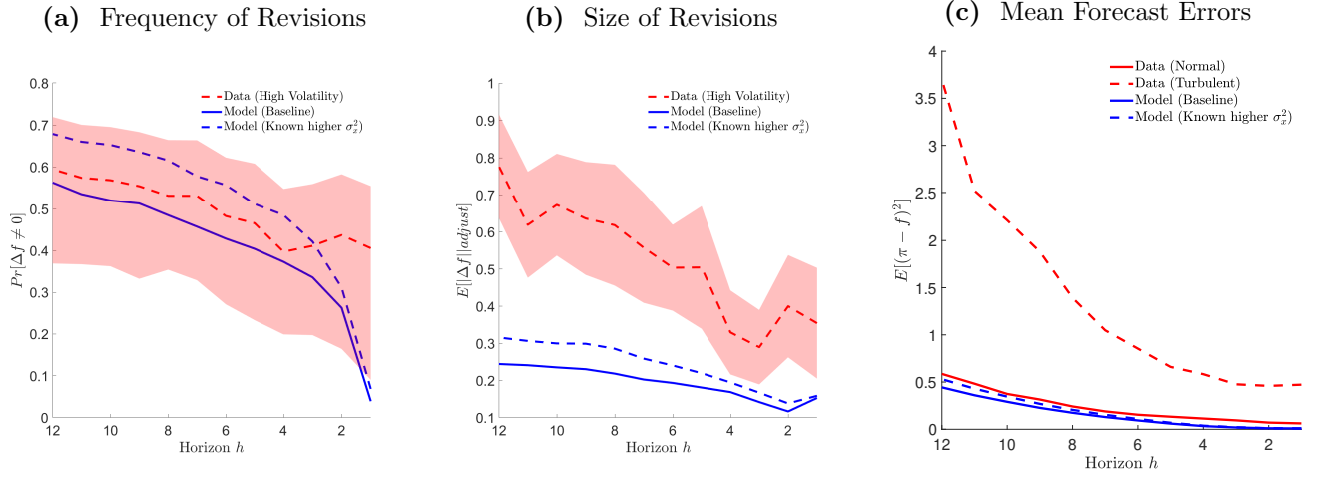
The comparison is organized around three informational regimes. In the baseline learning specification, forecasters infer regime changes over time. In the known-volatility counterfactual, forecasters have full knowledge of the inflation process and therefore adjust optimally as soon as the regime changes. In the undisclosed-volatility counterfactual, the true volatility changes but forecasters continue to behave as if the economy remained in the tranquil state. Comparing these three cases allows us to separate the role of uncertainty about the inflation process from the endogenous response of revision policies.

5.1 Counterfactual I: Known change in volatility

We first consider a full-information benchmark in which forecasters immediately observe the volatility regime and adjust optimally when inflation uncertainty rises. Relative to the baseline learning model, this counterfactual shuts down uncertainty about the state itself and isolates the effect of a more volatile environment on optimal revision behavior.

Figure XVI compares this counterfactual to the data and to the baseline model. In the data, moving from tranquil to volatile periods raises both the frequency and the size of revisions. The model reproduces these directional patterns. When forecasters know that volatility has increased, they revise more often and by larger amounts because the distribution of future inflation becomes more dispersed and the probability of crossing the inaction threshold rises. Forecast errors also increase, although less than in the data. This comparison shows that changes in optimal adjustment induced by higher volatility can account for an important part of the state dependence in revisions, but not all of the deterioration in forecast accuracy.

Figure XVI – Known changes in volatility



Notes: Red lines are from Bloomberg data. Tranquil = 2010-20. Turbulent = 2008-2009 and 2021-23. Blue lines use model-simulated data.

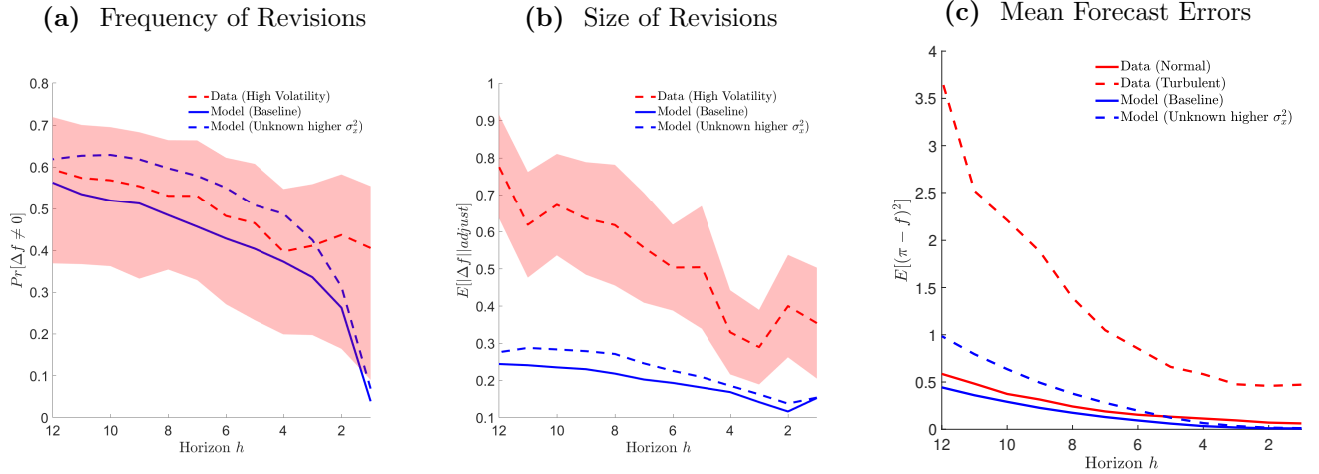
5.2 Counterfactual II: Undisclosed change in volatility

We next consider the opposite extreme: volatility rises in the true inflation process, but forecasters do not perceive the regime change and continue to behave as if the economy remained in the tranquil state. In this case, agents face larger shocks, but their revision policies remain fixed. Relative to the baseline, this counterfactual isolates the pure uncertainty effect of higher volatility, abstracting from endogenous changes in adjustment behavior.

Figure XVII shows that, under this counterfactual, forecast errors rise sharply because the environment becomes harder to predict while forecasters fail to adapt. By contrast, the response of revision frequency and revision size is much more muted than in the baseline learning model, precisely because agents do not re-optimize in response to the change in volatility. Comparing this case to the baseline clarifies the decomposition: part of the deterioration in forecast performance during volatile periods reflects higher uncertainty per se, whereas the increase in revision activity depends on forecasters recognizing that the environment has changed and adjusting their policies accordingly.

Interpretation. Taken together, these exercises clarify the mechanism behind state-dependent forecasting. The baseline learning model combines two effects: higher volatility makes inflation more difficult to forecast, and learning about the new regime changes the incentives to revise in the presence of fixed adjustment costs and strategic concerns. The known-volatility counterfactual isolates the policy effect of higher volatility under full information, while the undisclosed-volatility counterfactual isolates the pure uncertainty effect when agents fail to adjust their policies. The comparison across the three cases therefore shows how much of state dependence comes from uncertainty itself and how much comes from the endogenous response generated by κ and r .

Figure XVII – Unknown changes in volatility



Notes: Red lines are from Bloomberg data. Tranquil = 2010-20. Turbulent = 2008-2009 and 2021-23. Blue lines use model-simulated data.

6 Do Large Shocks Travel Fast? (In progress)

This exercise studies how large inflation shocks affect forecast revisions. The motivation stems from state-dependent models of price adjustment, in which sufficiently large shocks move firms out of their inaction regions and trigger immediate responses (Cavallo, Lippi and Miyahara, 2024). We ask whether an analogous mechanism operates in the formation of expectations.

The key idea is that, beyond changes in the volatility regime, unusually large inflation realizations may directly push forecasters outside their inaction regions, leading to rapid and widespread revisions. To study this channel, we feed the model with the actual sequence of inflation realizations and compare the implied forecast dynamics with those observed in the data.

If the model captures the mechanics of lumpy forecast adjustment well, large shocks should generate a highly nonlinear response: revisions should become both more likely and larger precisely when inflation realizations are unusually far from what forecasters had expected. In that sense, large shocks should *travel fast* through the forecasting process, just as they do in state-dependent models of price setting.

This exercise is useful for two reasons. First, it provides a direct test of the inaction mechanism at the level of expectations, rather than through changes in average volatility alone. Second, it helps distinguish between two sources of forecast adjustment: gradual learning about a changing environment and immediate responses to unusually large realizations. Comparing individual and consensus forecasts in these episodes allows us to assess whether expectation formation exhibits the same type of nonlinear propagation emphasized in the price-setting literature.

7 Conclusions

Our findings have direct implications for monetary policy. Standard models often assume that expectations adjust smoothly to shocks, but our results suggest that the responsiveness of expectations is itself state-dependent. In volatile periods, forecasters revise more frequently and by larger amounts, making expectations more reactive to new information. At the same time, forecasts become less precise, less anchored to the consensus, and more dispersed. This combination makes communication and stabilization more difficult, precisely when anchoring expectations is most important. More broadly, our results suggest that macroeconomic models of expectation formation should explicitly incorporate volatility and learning about regimes, since uncertainty changes how inflation shocks propagate through beliefs.

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